# Memo

2450 Venture Oaks Way. Sacramento Ca. 95833 (916) 648-9700 - FAX (916) 648-6534

**Date:** 22 July 2024

**To:** Maria Zupo
Kitchell CEM, Inc., Project Manager II

**From:** Alan Beck
Kitchell CEM, Inc., Project Scheduler

**Regarding:** 6691 Merritt College Horticulture –June 2024 Schedule Review

## Purpose of Memo

Kitchell CEM, Inc. ("Kitchell") has prepared a technical analysis of the June 2024 schedule update for the Merritt College Horticulture project. The schedule file is titled "446 – MERRITT-12.xer", received by Kitchell on 17 July 2024. This memo communicates the analysis and review of the schedule file. The schedule was reviewed for compliance based on general scheduling best practices**.**

## Schedule Disposition

The scheduled substantial completion date is **12/11/24** indicating a late finish. The BEI is lower than desired at **0.72** and the SPI is very low at **0.52**. The critical path is through **Roof Framing** & **Roof Sheathing**.

## Executive Summary

**Project Milestones**

Contractual NTP was **02/06/23**, substantial completion at **8/8/24**, and final completion at **9/6/24.** TIA#2 has added 70 calendar days and pushed the completion date to **11/16/24**.

| Contractual Milestone/ Variance Tracking | | | | | | |
|---|---|---|---|---|---|---|
| Activity ID | Contractual Milestones | Adjusted Contract | May-24 | Jun-24 | Variance from Contract | Variance from Previous Month |
| PM 1000 | NTP | 2/6/2023 | 2/6/2023 | 2/6/2023 | 0 | 0 |
| PM 1030 | All Submittals Provided | 5/7/2023 | 3/11/2024 | 3/11/2024 | -309 | 0 |
| PM 1040 | Slab on Grade Complete | 8/21/2023 | 11/16/2023 | 11/16/2023 | -87 | 0 |
| PM 1050 | Building Weather Tight | 3/12/2024 | 6/26/2024 | 6/26/2024 | -106 | 0 |
| PM 1060 | Substantial Completion | 10/17/2024 | 11/4/2024 | 12/11/2024 | -55 | -37 |
| PM 1010 | Final Project Completion | 11/16/2024 | 12/17/2024 | 1/23/2025 | -68 | -37 |

**Current Variance**

The final project completion milestone is **1/23/25** indicating they are **68-days behind of schedule**.

**Milestone Variance**

There is a substantial delay in getting the Library weather tight, and a delay in the final completion of the project.

> ***TIA 02 approved, extending duration of the project to 648 days and Final Completion is now 16-Nov-2024.**

## Project Health

Project health will be assessed using earned value and earned schedule metrics through the current payment application. Using an estimated planned value, and the Contractors earned actual value(s), the table below tabulates the current project status.


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| Original Contract | Change Orders | Current Contract | Completed Work | Remaining Work | % Budget Complete | % Duration Complete |
|---|---|---|---|---|---|---|
| $24,604,999.90 | $113,967.85 | $24,718,967.75 | $10,181,000.14 | $14,537,967.61 | 41% | 88% |
| Schedule Performance (Value) | | A = Current Earned Value | | B = Current Planned Value | | SPI (v) = A/B |
| **SPI (v)** | | $11,662,499.77 | | $22,318,515.32 | | 0.52 |
| Earned Schedule | | Current Actual Time | | Current Earned Date | | Schedule Variance (days) A-B |
| **ES** | | 30-Jun-24 | | 12-Dec-23 | | (201) |
| Schedule Performance (Time) | | A = Earned Schedule | | B = Actual Time | | SPI (t) =A/B |
| **SPI (t)** | | 309 | | 510 | | 0.61 |

- The Schedule Performance Index **SPI(v)=0.52** shows the ratio of the value of worked performed to work planned and is currently at **52%**.
- The earned schedule date, the point in time where the project is from a performance perspective, is 12-Dec-23 (201-days behind).
- Contractor cash flow needs to be discussed and possibly recalibrated.

## Contractor Narrative

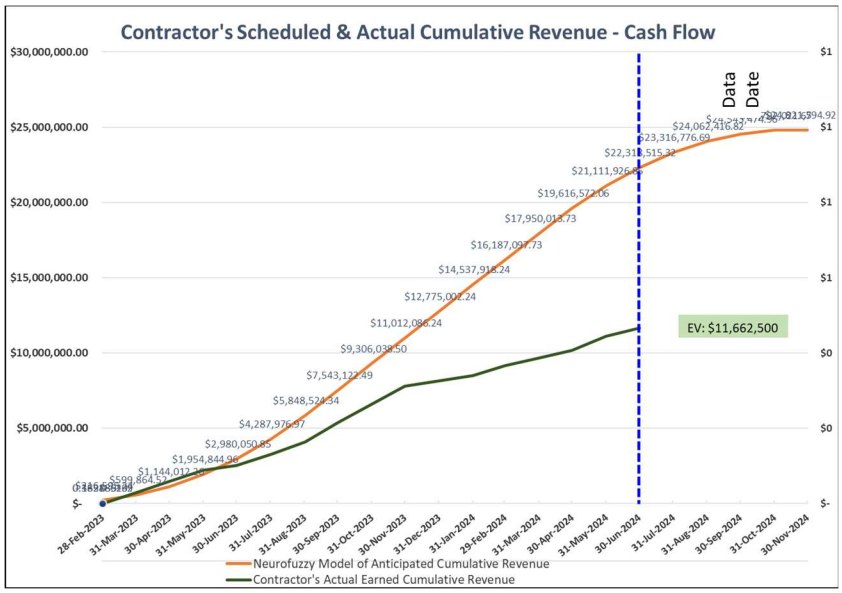
The following is a general synopsis of the Contractor's monthly report.

- Contractor states CCDs 23-26 are the causes of delays for the project this month.

## Project Health Analysis

### Schedule Performance

The graph illustrates the work plan cash flow over the life of the project, as provided by the Contractor. As the project progresses each month, earned value metrics can be applied using the planned curve and the payment application. This is for the purpose of assessing schedule performance and to forecast a reasonable completion date, if necessary. This methodology is crucial to understanding progress aside from milestone variance and activity execution.




- **The current earned schedule is tracking behind the planned schedule by 201-days.**



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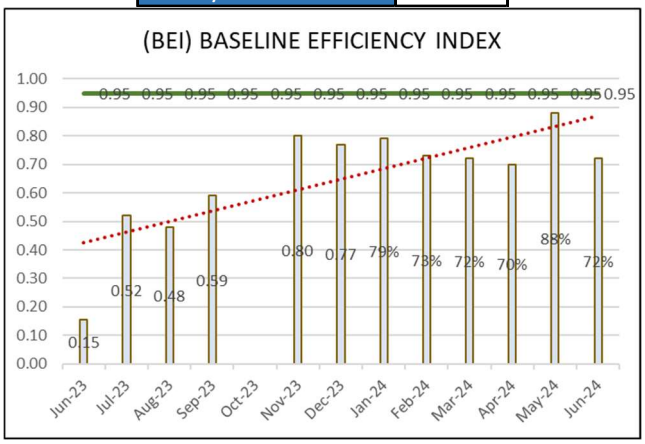
**Baseline Execution Index (BEI)**

The Baseline Execution Index (**BEI**) is used to track and report how closely a project is being executed compared to the planned execution within a specific time period. The BEI is a ratio of the overall number of activities completed compared to the number of activities that were planned to finish by the data date. If the Contractor is completing more tasks than planned, the BEI will be greater than 1.0. The target for this metric is to be **greater than or equal to 0.95**.

| BEI | |
|---|---|
| A: Actual Finish | 289 |
| B: Planned Finish | 400 |
| BEI: A/B | 72% |




- The schedule's cumulative BEI is at **0.72**. meaning that 72% of activities that were planned to be complete were in fact completed. This is lower than the desired rate of 0.95, but it is an improvement over the previous months.

**Schedule Relevance Index (SRI)**

The **SRI** is the measurement of activities planned for this month compared to how many actually started/ completed. This metric monitors the schedule relevance to work-in-place and offers insight to how well the Contractor weekly lookahead schedule adheres to the schedule.

| SRI Finishes | |
|---|---|
| A - Actual Finish | 10 |
| B - Planned Finish | 121 |
| SRI (A/B) | 0.08 |
| **SRI Starts** | |
| A - Actual Start | 17 |
| B - Planned Start | 103 |
| SRI (A/B) | 0.17 |

- 8% of the planned activities were finished this month.
- 17% of the planned activities were started this month.
- These values are significantly low and does not bode well for an on-time finish.

**Total Float Distribution Trend**

The total float analysis illustrates the float trend of outstanding activities from different risk levels. This tool is used to understand the migration of activities total float toward critical. Total float is amount of time an activity can be delayed without delaying the overall project duration. The Float distribution shows a trend toward high and low float with no moderate float.



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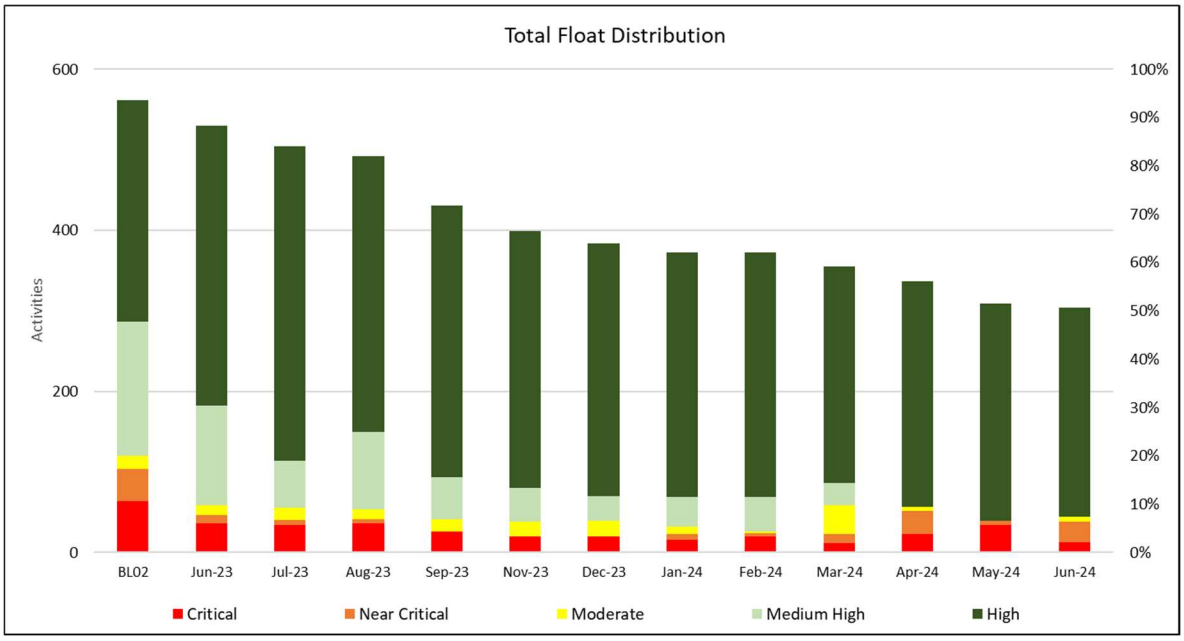


- High Float is at 86%.
- Near-Critical and Critical Float is at 12% much lower than the best practice of 25%.

**Critical Path**

The longest critical path is generally described through the library as follows:

- **Roof Framing** through **7/12/2024.**
- **Exterior Envelope** through **7/23/2024**.
- **Interior Buildout** through **10/31/2024**.
- **MEP** through **11/19/2024**.
- **Final Project Completion 1/23/2025**.

## Critical Analysis

**Critical Path Variance**

The critical path is still driven by the extended duration of **Roof Sheathing** and **Roof Framing.**

**Multiple Float Paths**

Activities within 10-days of the critical path are near-critical, as they have the potential of becoming critical with even slight delays. Near-critical activities that become critical can add to or change the entire workflow of the critical path. Currently there are 10 multiple float paths for the project that are shown to affect the longest critical path this is likely to change as the project continues.

*See attachments for detailed critical, variance and near critical activity reports.

**Analysis**

- The low values for SPI shows that the project is not where the project was projected to be currently. An increase in productivity and close adherence to the schedule would help to have the project end on time.

## Technical Analysis

The following table is derived from the attached forensic and logic analysis reports.



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**Schedule Diagnostic**

| Best Practice | Occurrences | Issue / Specification | Action |
|---|---|---|---|
| **Source** | | **Schedule Diagnostic** | |
| Activities with Finish-Start + Lag relationships | 0 | Can hide float | Ok |
| Activities with Start-Finish relationships | 0 | Not widely accepted as standard | Ok |
| Activities with Open Ends | 20 | Can skew total float and critical path<br>NTP and Final Completion are acceptable | Review & Revise |
| Out-of-Sequence | 8 | Can skew critical path | Review & Revise |
| Negative Lag | 0 | Can be better represented with alternate relationship | Ok |

| Source | | Schedule Comparison | |
|---|---|---|---|
| Best Practice | Occurrences | Issue / Specification | Action |
| Activities (Added/Modified/Deleted) | 12/ 0 /0 | Contractor to provide an explanation | |
| Relationship (Added/Modified/Deleted) | 54 / 0 / 22 | Contractor to provide an explanation | |
| Original Duration Changes (Increased/Decreased) | 0 / 0 | Contractor to provide an explanation | |

*See attachments for Schedule Diagnostic and Schedule Comparison

## Action Items

1. Contractor to address Critical Path change.

## Attachments

1. Remaining Activities
2. Variance Report
3. Longest Critical Path
4. Longest Critical Path - Alternative
5. Multiple Float Paths
6. Multiple Float Path Variance
7. Work Completed
8. Work Planned 30
9. Work Planned 60
10. Work Planned 90
11. Schedule Diagnostic
12. Schedule Comparison

## Disclaimer

The findings, observations, and/or opinions expressed in this review are not intended to direct the Contractor to revise its intended methodology or act as written authorization to amend any part of any of the Contract documents. Any questions or comments regarding this schedule review can be directed to Kitchell's project manager. Additionally, the review and rejection or non-rejection of any schedule, or any part of a schedule shall not be construed as a complete review and analysis of the submitted schedule(s). The acceptance of a schedule or not rejecting any part of it does not relieve the Contractor from any requirements in the Contract, nor is the acceptance an authorization to change any terms or conditions of the underlying Contract Specifications without the Owner's written authorization. The Contractor is respectfully reminded that failure of the Contractor to include any element of the Work within the schedule or any inaccuracy in the schedule will not relieve responsibility for accomplishing the Work in accordance with the Contract documents and within the Contract time. The Contractor is solely responsible for ensuring that all subcontractors and suppliers comply with the requirements of the Contract schedule for their portions of the Work, and that all schedules submitted to the Owner have been developed after coordination with subcontractors and suppliers. If the Contractor believes they are entitled to a Contract time extension, they are respectfully reminded that they are obligated to prepare and submit a time extension request that is consistent with Contract requirements to maintain and established entitlement. Failure to provide the necessary items and documentation may result in the automatic waiver by the Contractor for an extension and may also result in the Contractor in having to recovery or mitigate any loss time at its own cost.



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